Theoretical problems Fadone ① (not all)

(4) $\boxed{\phi(x) = Ax+b \to \phi^{-1} = ?}$

Proof.

ϕ = transformation $\Leftrightarrow$ $\det A \neq 0$, A inv. $\Leftrightarrow$ $\exists\, \phi^{-1}$.

$\phi(x) = \underbrace{Ax+b = y}$ $\Rightarrow$ $Ax = y-b \;/\cdot A^{-1}$

$x = A^{-1}\cdot(y-b) = \phi^{-1}(y)$

$\Rightarrow$ $\underline{\phi^{-1}(x) = A^{-1}x - A^{-1}b}$

lin. part: A^{-1}

transl. part: $-A^{-1}b$

(5) $\boxed{\phi(x) = Ax+b \text{ isom} \Leftrightarrow A^{-1} = A^T}$

Isometry = map that preserves distances. $d(\phi(P), \phi(Q)) = d(P,Q)$

Proof.

$d(\phi(P), \phi(Q)) = |(AP+b)-(AQ+b)| = |A\underbrace{(P-Q)}_{u}|$

$d(P,Q) = P-Q$

isom $\Leftrightarrow$ $AP+\cancel{b} - AQ - \cancel{b} = P-Q$ $\Rightarrow$ $A(P-Q) = P-Q$

"$\Rightarrow$" we know: ϕ isom.

$P-Q = u \neq 0$ (distance)

$|Au| = |u|\;|^2$ $\Rightarrow$ $\cancel{A^2u^2 = u^2}$ $\cancel{\Rightarrow A^2u^2 - u^2 = 0 \Rightarrow}$

$\cancel{\Rightarrow u^2(A-I)=0}$

$|Au|^2 = |u|^2$ $\Rightarrow$ $\langle Au, Au\rangle = \langle u,u\rangle$

$\Rightarrow (Au)^T.Au = u^Tu$ $\Rightarrow$ $u^TA^T.Au = u^Tu$ $\Rightarrow$

$\Rightarrow \underline{u^T}A^T.Au - \underline{u^T}u = 0$

$\Rightarrow u^T(A^T.Au - u) = 0$

$\Rightarrow \underbrace{u^T}_{\neq 0}.(\underbrace{u}_{\neq 0}\underbrace{(A^TA - I)}) = 0$

$\Leftrightarrow A^TA = I \Rightarrow A^{-1} = A^T.$

"$\Leftarrow$" we know: $A^{-1} = A^T \Rightarrow A^TA = I$

$(\forall)P,Q \Rightarrow d(\phi(P), \phi(Q)) = |A(P-Q)|\;|^2$

$\Rightarrow \|A(P-Q)\|^2 = (A(P-Q))^T.(A(P-Q)) = (P-Q)^T\underbrace{A^T.A}_{I}(P-Q) = (P-Q)^T(P-Q)$

$v^Tv = \|v\|^2 = \langle v,v\rangle$ $|P-Q|^2 = d(P,Q) \Rightarrow \phi$ isometry.

(7) $\boxed{A \in SO(2) \iff A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \theta \in \mathbb{R}}$

$SO(2) = \{ A \in M_2(\mathbb{R}) : \underbrace{A^TA = I}_{\text{orthogonal matrix}}, \underbrace{\det A = 1}_{\text{preserves orientation}} \}$

$\begin{cases} a^2+b^2=1 \\ b^2+d^2=1 \end{cases} \Rightarrow a = \cos\theta$; $ac+bd=0$, $ad-bc=1$

$A^TA = \begin{bmatrix} a^2+c^2 & ab+cd \\ ab+cd & b^2+d^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Proof.

"$\Rightarrow$" We know: $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $A^TA = I$, $\det A = 1$.

$\begin{cases} a^2+c^2 = 1 \Rightarrow a = \cos\theta, c = \\ b^2+d^2=1 \Rightarrow d = \cos(\tilde\theta), b \\ ab+cd=0 \\ ad+bc=1 \end{cases}$

$a,b,c,d = ?$

$AA^T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} a^2+b^2 & ac+bd \\ ac+bd & c^2+d^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

BUT IT SHOULD HAVE BEEN A^TA!

$0 = \cos\theta \sin\tilde\theta + \sin\theta\cos\tilde\theta$
$0 = \sin(\theta+\tilde\theta)$

$ad-bc=1.$

$1 = \cos\theta\cos\tilde\theta + \sin\tilde\theta\sin\theta$
$1 = \cos(\theta+\tilde\theta)$

$\Rightarrow \begin{cases} a^2+b^2=1 \Rightarrow a^2 = 1-b^2 \\ c^2+d^2=1 \Rightarrow d^2 = 1-c^2 \\ ac+bd=0 \Rightarrow ac = -bd \mid^2 \Rightarrow a^2c^2 = b^2d^2 \\ ad-bc=1 \end{cases}$

$\Rightarrow \tilde\theta = -\theta + 2k\pi, k \in$

$\Rightarrow (1-b^2)c^2 = b^2(1-c^2)$

$\Rightarrow c^2 - \cancel{b^2c^2} = b^2 - \cancel{b^2c^2} \Rightarrow b^2 = c^2$

$\Rightarrow a^2 = d^2.$

$ac = -bd$

$ad-bc = 1 \mid^2 \Rightarrow a^2d^2 - 2abcd + b^2c^2 = 1$

$abcd = ac \cdot bd = -bd \cdot bd = -b^2d^2$

$\Rightarrow d^4 - 2b^2d^2 + b^4 = 1$

$\Rightarrow (d^2-b^2) = 1 \Rightarrow b^2 = d^2.$

orthonormal columns $\Rightarrow$ first col: $\boxed{a^2+c^2=1} \Rightarrow \begin{pmatrix} \cos \\ \sin \end{pmatrix}$

$\Rightarrow$ $\boxed{\text{second col} \perp \text{first col}} \Rightarrow \begin{pmatrix} -\sin \\ \cos \end{pmatrix}, \begin{pmatrix} \sin \\ -\cos \end{pmatrix}$

$\det A = 1 \Rightarrow$ preserve orientation $\Rightarrow \begin{pmatrix} -\sin \\ \cos \end{pmatrix}$

$\Rightarrow A_1 = \begin{pmatrix} \cos & -\sin \\ \sin & \cos \end{pmatrix}$, $A_2 = \begin{pmatrix} \cos & \sin \\ \sin & -\cos \end{pmatrix}$

$\hookrightarrow \cos^2 + \sin^2 = 1$
$\det A_1$ //

$\hookrightarrow -\cos^2 - \sin^2 = -1$
$\det A_2$ //

But $\det A = 1$

$\Rightarrow$ choose $A_1 = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

STPPR

"$\Longleftarrow$"

We know: $A = \begin{pmatrix} \cos & -\sin \\ \sin & \cos \end{pmatrix}$

$A^T A = \begin{pmatrix} \cos & \sin \\ -\sin & \cos \end{pmatrix}\begin{pmatrix} \cos & -\sin \\ \sin & \cos \end{pmatrix} = \begin{pmatrix} \cos^2+\sin^2 & -\cos\sin+\sin\cos \\ -\cos\sin+\sin\cos & +\sin^2+\cos^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & +1 \end{pmatrix}$

$\det A = \begin{vmatrix} \cos & -\sin \\ \sin & \cos \end{vmatrix} = 1.$

$\Rightarrow A \in SO(2)$

⑧. $\boxed{\text{dir. isometry of } \mathbb{E}^2 \text{ that fixes a point} \Rightarrow \text{Id or Rot.}}$

Proof

$\Phi(x) = Ax + b$

dir. isometry $\Rightarrow$ $A^T A = I$, $\det A = 1$.

$\Rightarrow A \in SO(2) \Rightarrow A = \begin{bmatrix} \cos & -\sin \\ \sin & \cos \end{bmatrix}$

Fix point P: $\Phi(P) = P$

$\Rightarrow AP + b = P \Rightarrow$ ~~$AP - P =$~~

$\Rightarrow b = P - AP \Rightarrow b = P(I-A)$

$\Phi(x) = Ax + b = Ax + P(I-A) = Ax + P - AP = P + A(x-P)$

$\Rightarrow$ ROTATION centered at P.

• Partic. case: $\theta = 0$ rot. angle $\Rightarrow A = \begin{bmatrix} \cos 0 & -\sin 0 \\ \sin 0 & \cos 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$

$\Rightarrow \Phi(x) = I_2 \cdot x + b$

Is $b \neq 0$? $b = P(I-A) = P(I_2 - I_2) = 0$

$\cancel{\Phi(x) = P + I_2(x-P)} = I_2 \cdot x = x \Rightarrow \cancel{b=0}$.

$\left.\begin{array}{l} A = I_2 \\ b = 0 \end{array}\right\} \Rightarrow \Phi$ identity.

⑨. $\boxed{\text{composition of rotations} \Rightarrow \text{translation}}$

$Rot_{\theta_1, c_1} = f$

$Rot_{\theta_2, c_2} = g$

$Rot_{\theta, c}(x) = c + R_\theta(x - c)$ | gen. form of a rot.

$\boxed{= \underbrace{R_\theta x}_{\text{lin. part}} + \underbrace{(c - R_\theta c)}_{\text{transl. part}}}$

$$\begin{cases} f(x) = R_{\theta_1}x + (c_1 - R_{\theta_1}c_1) = R_{\theta_1}x + b_1 \\ g(x) = R_{\theta_2}x + (c_2 - R_{\theta_2}c_2) = R_{\theta_2}x + b_2 \end{cases}$$

$$(g\circ f)(x) = g(f(x)) = R_{\theta_2}(R_{\theta_1}x + b_1) + b_2 = \underbrace{R_{\theta_2}R_{\theta_1}}_{\text{lin. part}}x + R_{\theta_2}b_1 + b_2$$

$$\boxed{R_{\theta_2}R_{\theta_1} = R_{\theta_2+\theta_1}}$$

$\boxed{\text{Translation } x \mapsto x+t,\ \text{lin. part} = I_2}$

$g\circ f$ translation $\Leftrightarrow$ lin. part $= I_2$

$\Leftrightarrow R_{\theta_2}R_{\theta_1} = I_2$

$\Leftrightarrow R_{\theta_2+\theta_1} = I_2$

$\Leftrightarrow \underline{\theta_1 + \theta_2 = 2k\pi}$

Trivial translation $\to$ when transl vector $=0$ $\Rightarrow$ composition = identity

(10). $\boxed{\text{Homog. matrix of a rotation with } \theta \text{ and center } C}$

$\boxed{Rot_{\theta,c}(x) = c + R_\theta(x-c)} = R_\theta x + (c - R_\theta c) =$

$$= \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} + \left(\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} - \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}\right)$$

$$= \dots + \left(\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} - \begin{bmatrix} \cos\theta \cdot c_1 - c_2 \cdot \sin\theta \\ c_1 \cdot \sin\theta + c_2 \cdot \cos\theta \end{bmatrix}\right) = \dots + \overbrace{\begin{bmatrix} c_1(1-\cos\theta) + c_2 \cdot \sin\theta \\ c_2(1-\cos\theta) - c_1 \cdot \sin\theta \end{bmatrix}}^{b}$$

Homog. matrix (general): $\phi(x) = Ax + b \mapsto [Q] = \begin{bmatrix} A & b \\ 0 & 1 \end{bmatrix}$

Here: $A = R_\theta$

$b = c - R_\theta c$

Homog. matrix: $\begin{bmatrix} \cos\theta & -\sin\theta & c_1(1-\cos\theta) + c_2 \cdot \sin\theta \\ \sin\theta & \cos\theta & c_2(1-\cos\theta) - c_1 \cdot \sin\theta \\ 0 & 0 & 1 \end{bmatrix}$

3) $\boxed{A \in SO(3) \text{ admits } 1 \text{ as eigenval.}}$

$A \in SO(3) \Rightarrow \{A \in \mathcal{M}_3(\mathbb{R}) : A^T A = I_3,\ \det A = 1\}$

$\lambda = 1 \Rightarrow \det(A - 1 \cdot I_3) = 0.$

$\boxed{\det A = \det A^T = 1}$

$\boxed{\det(A \cdot B) = \det A \cdot \det B}$

$\det(A - I_3) = 0 \ | \cdot \det(A^T) = 1$

$\Rightarrow \det(A^T) \cdot \det(A - I_3) = 0$

$(\Leftrightarrow) \det(A^T(A - I_3)) = 0$

$(\Leftrightarrow) \det(A^T A - A^T \cdot I_3) = 0$

$(\Leftrightarrow) \det(I_3 - A^T) = 0$

$(\Leftrightarrow) \det(I_3 - A)^T = 0$

$(\Leftrightarrow) \begin{cases} \det(I_3 - A) = 0 \ (\Leftrightarrow) \det(-(A - I_3)) = 0 \ (\Leftrightarrow) -\det(A - I_3) = 0 \\ \det(A - I_3) = 0 \end{cases}$

$\Rightarrow \det(A - I_3) = -\det(A - I_3)$

$\Rightarrow 2\det(A - I_3) = 0 \Rightarrow \det(A - I_3) = 0$, indeed

$\Rightarrow \lambda = 1$ eigenvalue.

(14) $\boxed{A \in O(3) \text{ with } \det A = -1 \text{ admits } -1 \text{ as eigenval.}}$

$\lambda = -1$ eigenvalue $\Rightarrow \det(A + I_3) = 0.$

$\boxed{\det(A) = \det(A^T) = -1}$

$\det(A + I_3) = 0 \ | \cdot \det(A^T)$

$\Rightarrow \det(A^T) \cdot \det(A + I_3) = 0$

$\Rightarrow \det(A^T(A + I_3)) = 0$

$\Rightarrow \det(\underbrace{A^T A}_{I} + A^T \cdot I_3) = 0$

$\Rightarrow \det(I + A^T) = 0$

$\Rightarrow \det(I + A)^T = 0$

$\Rightarrow \begin{cases} \det(I + A) = 0 \\ -\det(I + A) = 0 \end{cases}$

$\Rightarrow 2\det(I + A) = 0 \Rightarrow \det(I + A) = 0$

$\Rightarrow \lambda = -1$ eigen value

$= \mid I_m - v_n T$

(16) **Deduce can. eq. of the ellipse**

$\mathcal{E}: \{P \in \mathbb{E}^2 \mid d(P,F_1) + d(P,F_2) = 2a\}, \ (\forall)\, a > 0$

$F_1(c,0),\ F_2(-c,0),\ c>0,\ P(x,y)$

$$\mathcal{E}: \sqrt{(x-c)^2+(y-0)^2} + \sqrt{(x+c)^2+(y-0)^2} = 2a$$

$$\Rightarrow \sqrt{(x-c)^2+y^2} - 2a = -\sqrt{(x+c)^2+y^2} \quad |^2$$

$$\Rightarrow (x-c)^2+\cancel{y^2} - 4a\sqrt{(x-c)^2+y^2} + 4a^2 = (x+c)^2+\cancel{y^2}$$

$$\Rightarrow -2xc - 4a\sqrt{\dots} + 4a^2 = 2xc$$

$$\Rightarrow 4a^2 - 4a\sqrt{(x-c)^2+y^2} - 4xc = 0 \quad |:4$$

$$\Rightarrow a^2 - cx = a\sqrt{(x-c)^2+y^2} \quad |^2$$

$$\Rightarrow a^4 - 2a^2cx + c^2x^2 = a^2(x^2-2cx+c^2+y^2)$$

$$\Rightarrow a^4 - \cancel{2a^2cx} + c^2x^2 = a^2x^2 - \cancel{2a^2cx} + a^2c^2 + a^2y^2 \quad |:a^2$$

$$\Rightarrow a^2 + \frac{c^2x^2}{a^2} = x^2+y^2+c^2$$

$$\Rightarrow \underbrace{a^2-c^2}_{b^2} = x^2+y^2-\frac{c^2x^2}{a^2} \quad |:b^2$$

$$\Rightarrow 1 = \frac{x^2}{b^2} + \frac{y^2}{b^2} - \frac{c^2x^2}{b^2a^2}$$

$$\Rightarrow 1 = \frac{x^2}{b^2}\left(\frac{a^2}{1}-\frac{c^2}{a^2}\right) + \frac{y^2}{b^2}$$

$$\Rightarrow 1 = \frac{x^2}{\cancel{b^2}} \cdot \frac{\cancel{a^2-c^2}}{a^2} + \frac{y^2}{b^2}$$

$$\Rightarrow 1 = \frac{x^2}{a^2} + \frac{y^2}{b^2} : \mathcal{E}$$

17) $\boxed{\mathcal{E}\ \frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \quad \text{ELLIPSE} \to \text{tg lines of given slope}}$

$$\begin{cases} \mathcal{E}: \frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \\ \ell: kx+m=y \end{cases} \Rightarrow \dots\dots, \ \Delta=0 \Rightarrow a^2k^2+b^2-m^2=0$$

$$\Rightarrow m^2=a^2k^2+b^2$$

$$\Rightarrow m=\pm\sqrt{a^2k^2+b^2}$$

$$y=kx+m \Rightarrow \boxed{y=kx\pm\sqrt{a^2k^2+b^2}}$$

18) $\boxed{\mathcal{E}: \frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \quad \text{ELLIPSE} \to \text{tg lines at given point } (x_0,y_0)}$

$$\begin{cases} \mathcal{E}: \frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \\ \ell: \begin{cases} x=x_0+t\cdot v_x \\ y=y_0+t\cdot v_y \end{cases} \end{cases}$$

ℓ tangent to $\mathcal{E} \Leftrightarrow |\ell\cap\mathcal{E}|=1$

$$\Rightarrow \dots \Rightarrow t^2(\dots)+2t\underbrace{(\dots)}_{\text{should be } 0}+\underbrace{\frac{x_0^2}{a^2}+\frac{y_0^2}{b^2}-1}_{0}=0$$

$$\frac{x_0v_x}{a^2}+\frac{y_0v_y}{b^2}$$

$n\left(\frac{x_0}{a^2},\frac{y_0}{b^2}\right)$ normal vector for ℓ.

$$\Rightarrow \frac{x_0}{a^2}(x-x_0)+\frac{y_0}{b^2}(y-y_0)=0$$

$$\Rightarrow \frac{xx_0}{a^2}+\frac{yy_0}{b^2}\underbrace{-\frac{x_0^2}{a^2}-\frac{y_0^2}{b^2}}_{1}=0$$

$$\Rightarrow T\ \mathcal{E}: \frac{xx_0}{a^2}+\frac{yy_0}{b^2}=1$$

19) $\boxed{\text{Tangent lines of an ELLIPSE are } \parallel}$

$P(x_0,y_0)$ one endpoint of diam.

$P'(-x_0,-y_0)$ other endpoint of diam.

Tangent at P: $T_{(x_0,y_0)}\mathcal{E}_{a,b}: \frac{xx_0}{a^2}+\frac{yy_0}{b^2}=1 \Rightarrow n_P\left(\frac{x_0}{a^2},\frac{y_0}{b^2}\right)$

Tangent at P': $T_{(x_0,y_0)}\mathcal{E}_{a,b}: -\frac{xx_0}{a^2}-\frac{yy_0}{b^2}=+1 \ /\cdot(-1)$ ~~$\Rightarrow n_{P'}\left(\frac{x_0}{a^2},\frac{y_0}{b^2}\right)$~~

$$\Rightarrow \frac{xx_0}{a^2}+\frac{yy_0}{b^2}=-1$$

$\boxed{n_P=n_{P'}}$ $\Rightarrow \parallel$ tangent lines. $\Rightarrow n_{P'}\left(\frac{x_0}{a^2},\frac{y_0}{b^2}\right)$

(20) $\mathcal{H}: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ HYPERBOLA, tg lines of a given slope

$$\begin{cases} \mathcal{H}: \\ y = kx + m \end{cases} \Rightarrow \dots \Rightarrow \Delta = 0 \Rightarrow m = \pm\sqrt{a^2k^2 - b^2}$$

$$\boxed{y = kx \pm \sqrt{a^2k^2 - b^2}}$$

(21) $\mathcal{H}: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ HYPERBOLA, tg lines at given point (x_0, y_0)

$$\begin{cases} \mathcal{H}: \dots \\ l: \begin{cases} x = x_0 + t \cdot v_x \\ y = y_0 + t \cdot v_y \end{cases} \end{cases}$$ l tangent to $\mathcal{H}$ $\Leftrightarrow$ $|l \cap \mathcal{H}| = 1$

$$\Rightarrow t^2(\dots) + 2t \cdot \underbrace{(\dots)}_{\text{should be } 0} + \underbrace{\frac{x_0^2}{a^2} - \frac{y_0^2}{b^2} - 1}_{0} = 0$$

$$\frac{x_0 v_x}{a^2} - \frac{y_0 v_y}{b^2} = 0$$

$\Rightarrow n\left(\frac{x_0}{a^2}, \frac{y_0}{b^2}\right)$ normal vector for line l.

$$\Rightarrow \frac{x_0}{a^2}(x - x_0) + \frac{y_0}{b^2}(y - y_0) = 0$$

$$\Rightarrow \frac{xx_0}{a^2} - \frac{yy_0}{b^2} - \underbrace{\left(\frac{x_0^2}{a^2} - \frac{y_0^2}{b^2}\right)}_{1} = 0$$

$$T_{..}\mathcal{H}: \frac{xx_0}{a^2} - \frac{yy_0}{b^2} = 1$$

(22) $\mathcal{P}: y^2 = 2px$, PARABOLA, tg line of given slope

$$\begin{cases} \mathcal{P}: \dots \\ y = kx + m \end{cases} \Rightarrow \dots \Delta = 0 \Rightarrow \boxed{y = kx \pm \frac{p}{2k}}$$

(23) $\mathcal{P}: y^2 = 2px$, tg at given point (x_0, y_0)

$$\begin{cases} \mathcal{P}: \\ l: \begin{cases} x = x_0 + t \cdot v_x \\ y = y_0 + t \cdot v_y \end{cases} \end{cases}$$

> $F(x,y) = 0$
> tg. at $P(x_0, y_0)$ has normal vector $\nabla F(x_0, y_0)$
> tg: $\nabla F(x_0, y_0) \cdot \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix} = 0$

$$F(x,y) = y^2 - 2px = 0$$

$H, v(P)$ _ / _ vector

(20) $\mathcal{H}: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ HYPERBOLA, tg lines of a given slope

$$\left\{\begin{array}{l} \mathcal{H}: \dots \\ y = kx + m \end{array}\right\} \Rightarrow \dots \Rightarrow \Delta = 0 \Rightarrow m = \pm\sqrt{a^2k^2 - b^2}$$

$$\boxed{y = kx \pm \sqrt{a^2k^2 - b^2}}$$

(21) $\mathcal{H}: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ HYPERBOLA, tg lines at given point (x_0, y_0)

$$\left\{\begin{array}{l} \mathcal{H}: \dots \\ l: \left\{\begin{array}{l} x = x_0 + t \cdot v_x \\ y = y_0 + t \cdot v_y \end{array}\right\} \end{array}\right.$$

l tangent to $\mathcal{H}$ $\Leftrightarrow$ $|l \cap \mathcal{H}| = 1$

$$\Rightarrow t^2(\dots) + 2t \cdot \underbrace{(\dots)}_{\text{should be } 0} + \underbrace{\frac{x_0^2}{a^2} - \frac{y_0^2}{b^2} - 1}_{0} = 0$$

$$\frac{x_0 \cdot v_x}{a^2} - \frac{y_0 \cdot v_y}{b^2} = 0$$

$\Rightarrow n\left(\frac{x_0}{a^2}, \frac{y_0}{b^2}\right)$ normal vector for line l.

$$\Rightarrow \frac{x_0}{a^2}(x - x_0) + \frac{y_0}{b^2}(y - y_0) = 0$$

$$\Rightarrow \frac{xx_0}{a^2} - \frac{yy_0}{b^2} - \underbrace{\left(\frac{x_0^2}{a^2} - \frac{y_0^2}{b^2}\right)}_{1} = 0$$

$$T_{\dots}\mathcal{H}: \frac{xx_0}{a^2} - \frac{yy_0}{b^2} = 1$$

(22) $\mathcal{P}: y^2 = 2px$, PARABOLA, tg line of given slope

$$\left\{\begin{array}{l} \mathcal{P}: \dots \\ y = kx + m \end{array}\right\} \Rightarrow \dots \Delta = 0 \Rightarrow \boxed{y = kx + \frac{p}{2k}}$$

(23) $\mathcal{P}: y^2 = 2px$, tg at given point (x_0, y_0)

$$\left\{\begin{array}{l} \mathcal{P}: \\ l: \left\{\begin{array}{l} x = x_0 + t \cdot v_x \\ y = y_0 + t \cdot v_y \end{array}\right\} \end{array}\right.$$

$F(x,y) = 0$

tg. at $P(x_0, y_0)$ has normal vector $\nabla F(x_0, y_0)$

tg: $\nabla F(x_0, y_0) \cdot \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix} = 0$

$$F(x,y) = y^2 - 2px = 0$$

$$\nabla F(x,y) = \left(\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}\right) = (-2p, 2y)$$

$$\nabla F(x_0, y_0) = (-2p, 2y_0)$$

$$(-2p, 2y_0)(x-x_0, y-y_0) = 0$$

$$\Rightarrow -2p(x-x_0) + 2y_0(y-y_0) = 0$$

$$\Rightarrow -p(x-x_0) + y_0(y-y_0) = 0$$

$$-px + px_0 + y_0 y - y_0^2 = 0$$

$$y_0 y = px - px_0 + y_0^2$$

$P \in$ parabola $\Rightarrow y_0^2 = 2px_0$

$$y_0 y = px - px_0 + 2px_0 = px + px_0 = p(x+x_0)$$

$$\boxed{yy_0 = p(x+x_0)}$$

(24) $\boxed{\mathcal{E}_{a,b,c}: \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \text{ ELLIPSOID, tg plane at } (x_0, y_0, z_0)}$

$$\begin{cases} \mathcal{E}: \dots \\ l: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_p \\ y_p \\ z_p \end{bmatrix} + t \cdot \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} \end{cases}$$

l tg. to $\mathcal{E}$ ~~$l \cap \mathcal{E}_{a,b,c}$~~ $\Leftrightarrow |l \cap \mathcal{E}| = 1$

$$\Rightarrow t^2(\dots) + 2t\underbrace{(\dots)}_{0} + \underbrace{\dots}_{1} = 1$$

Normal vector ..... $\Rightarrow \langle v, m \rangle = 0$

$$T_{\cdot}\mathcal{E} : \frac{x_p}{a^2}(x-x_p) + \frac{y_p}{b^2}(y-y_p) + \frac{z_p}{c^2}(z-z_p) = 0$$

......

$$\Rightarrow \frac{xx_p}{a^2} + \frac{yy_p}{b^2} + \frac{zz_p}{c^2} = 1$$

25 $C: \frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=0$ CONE, tg plane at (x_0, y_0, z_0)

~~$\begin{cases} C: \\ l: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \cdot \begin{bmatrix} x \end{bmatrix} \end{cases}$~~

Proof w Gradient

$$F(x,y,z)=\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}$$

$$\nabla F = \left(\frac{2x}{a^2}, \frac{2y}{b^2}, \frac{-2z}{c^2}\right)$$

$\nabla F(P): \left(\frac{2x_0}{a^2}, \dots\right)$ normal vector

$$\frac{2x_0}{a^2}(x-x_0)+\frac{2y_0}{b^2}(y-y_0)-\frac{2z_0}{c^2}(z-z_0)=0$$

$$\Rightarrow \frac{xx_0}{a^2}+\frac{yy_0}{b^2}-\frac{zz_0}{c^2}=0$$

26 $C: \frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=0$ ruled surface?

$$\begin{cases} C: \\ l: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \cdot \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} \end{cases}$$

$$\Rightarrow \frac{(t \cdot x_0)^2}{a^2}+\frac{(t \cdot y_0)^2}{b^2}-\frac{(t \cdot z_0)^2}{c^2}=0$$

$$\Rightarrow t^2 \underbrace{\left(\frac{x_0^2}{a^2}+\frac{y_0^2}{b^2}-\frac{z_0^2}{c^2}\right)}_{0}=0$$

$\Rightarrow$ l generator for the cone.

27 gradient

28 $H': \frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=1$ eq. of family I of generators

$$\frac{x^2}{a^2}-\frac{z^2}{c^2}=1-\frac{y^2}{b^2}$$

$$\left(\frac{x}{a}-\frac{z}{c}\right)\left(\frac{x}{a}+\frac{z}{c}\right)=\left(1-\frac{y}{b}\right)\left(1+\frac{y}{b}\right)$$

I: divide one way $\frac{\frac{x}{a}-\frac{z}{c}}{1-\frac{y}{b}}=\frac{1+\frac{y}{b}}{\frac{x}{a}+\frac{z}{c}}=\alpha$

$$\Rightarrow \begin{cases} \frac{x}{a} - \frac{z}{c} = \alpha\left(1 - \frac{y}{b}\right) \\ 1 + \frac{y}{b} = \alpha\left(\frac{x}{a} + \frac{z}{c}\right) \end{cases}$$

z) line.

(29) $\boxed{\mathcal{P}^h_{a,b}: \frac{x^2}{a} + \frac{y^2}{b} - 2z = 0, \text{ eq. tangent at point } (x_0, y_0, z_0)}$

$$\phi(x_0, y_0, z_0) = \left(\frac{x^2}{a}, \frac{y^2}{b}, -2z\right)$$

$$\nabla\phi = \left(\frac{2x}{a}, \frac{2y}{b}, -2\right) \text{ at } (x_0, y_0, z_0) \Rightarrow \left(\frac{2x_0}{a}, \frac{2y_0}{b}, -2z_0\right)$$

$$\frac{2x_0}{a}(x - x_0) + \frac{2y_0}{b}(y - y_0) - 2(z - z_0) = 0 \quad |:2$$

$$\frac{xx_0}{a} + \frac{yy_0}{b} + z - \left(\frac{x_0^2}{a} + \frac{y_0^2}{b} - z_0\right) = 0$$

$$\dots\dots - \underbrace{\left(\frac{x_0^2}{a} + \frac{y_0^2}{b} - 2z_0\right)}_{0} + z_0 = 0$$

$$T \dots \mathcal{P}^h_{a,b} = \frac{x x_0}{a} + \frac{y y_0}{b} + z - z_0 = 0.$$

(30) $\boxed{\mathcal{P}^h_{a,b}: \frac{x^2}{a} + \frac{y^2}{b} - 2z = 0, \text{ eq. of one of the families of gen.}}$

<u>Hyp par:</u>

$$\frac{x^2}{a} - \frac{y^2}{b} = 2z$$

$$\left(\frac{x}{a} - \frac{y}{b}\right)\left(\frac{x}{a} + \frac{y}{b}\right) = 2z \Leftrightarrow AB = 2z$$

$$\boxed{\mu B = \lambda} \Rightarrow B = \frac{\lambda}{\mu}$$

$$\Rightarrow A \cdot \frac{\lambda}{\mu} = 2z \Rightarrow \begin{cases} \lambda A = 2\mu z \\ \mu B = \lambda \end{cases} \Rightarrow \begin{cases} \lambda\left(\frac{x}{\sqrt{a}} - \frac{y}{\sqrt{b}}\right) = 2\mu z \\ \mu\left(\frac{x}{\sqrt{a}} + \frac{y}{\sqrt{b}}\right) = \lambda \end{cases}$$

Theoretical problems

$\boxed{\text{Problem 1}}$ $H: a_1x_1 + \dots + a_m x_m = 0$ hyperplane

$v(v_1, \dots, v_m)$ vector $\nparallel H$.

$a(a_1, \dots, a_m)$ vector

$$\boxed{Pr_{H,v}(P) = \left(I_m - \frac{v \cdot a^T}{v^T \cdot a}\right) \cdot P - \frac{a_{m+1}}{v^T \cdot a} v}$$ projection along v

$a = \begin{pmatrix} a_1 \\ \vdots \\ a_m \end{pmatrix}$, $P = \begin{pmatrix} p_1 \\ \vdots \\ p_m \end{pmatrix}$

Hyperplane equation: $a^T P + a_{m+1} = 0$.

Proof:

Let $P' = Pr_{H,v}(P)$

We project along direction v: $P' = P + t \cdot v$

$$\left.\begin{aligned} P' \in H: \quad & a^T P' + a_{m+1} = 0 \\ & P' = P + tv \end{aligned}\right\} \Rightarrow a^T(P + tv) + a_{m+1} = 0$$

$$a^T P + t \cdot a^T v + a_{m+1} = 0$$

$$\Rightarrow t = -\frac{a^T P + a_{m+1}}{a^T v}$$

$P' = P + t \cdot v$

$P' = P - \frac{a^T P + a_{m+1}}{a^T v} \cdot v$ $\qquad a^T v = \langle a, v \rangle = \langle v, a \rangle = v^T a.$

$P' = P - \frac{v a^T P}{v^T v} - \frac{a_{m+1}}{a^T v} \cdot v$

$\Rightarrow Pr_{H,v}(P) = \cancel{I_m} \; \cancel{I_m} \left(I_m - \frac{v a^T}{v^T a}\right) \cdot P - \frac{a_{m+1}}{v^T a} v$

$\boxed{\text{Problem 2}}$ π plane in $\mathbb{E}^3$ - passes through origin
- has normal vector m.

If $|m| = 1$, show: $\boxed{Pr_{\pi}^{\perp}(P) = (I_3 - m \otimes m) P}$

$\pi \subset \mathbb{E}^3$, $O \in \pi$, $m \perp \pi$, $|m| = 1$.

Proof:

Equation of a plane passing through origin with normal vector m: $m^T P = 0$.

Use $\boxed{\text{Problem 1}}$: $Pr_H^{\perp}(P) = \left(I_m - \frac{a \otimes a}{|a|^2}\right) P - \frac{a_{m+1}}{|a|^2} a \quad (1)$

$\to$ orthogonal projection onto a hyperplane = projection in the direction of a normal vector

General hyperplane eq: $a^T p + a_{m+1} = 0$

our hyperplane eq: $m^T p = 0$

$\Rightarrow a^T p + a_{m+1} = m^T p$

$\Rightarrow \begin{cases} m^T = a^T \\ a_{m+1} = 0 \end{cases} \Rightarrow m = a.$

Replace in formula (1): $Pr^{\perp}_{H}(P) = \left(I_m - \frac{m \otimes m}{\underbrace{|m|^2}_{|m|=1}} \right) P \Rightarrow$

$\Rightarrow Pr^{\perp}_{\pi}(P) = (I_m - m \otimes m) P.$

Problem 3 π plane in $\mathbb{E}^3$ - passing through origin
- normal vector m,

If $|m|=1$, show: $\boxed{Ref^{\perp}_{\pi}(P) = (I_3 - 2 \cdot m \otimes m) P.}$

$\pi \subset \mathbb{E}^3, 0 \in \pi, m \perp \pi, |m|=1.$

Proof:

$\boxed{Pr^{\perp}_{\pi}(P) = (I_3 - m \otimes m) P}$

$\boxed{(m \otimes m) P = \langle m, P \rangle m}$

$P = \underbrace{(I_3 - m \otimes m) P}_{\text{part in plane}} + \underbrace{(m \otimes m) P}_{\text{normal part}}$

after reflection: $Ref^{\perp}_{\pi}(P) = \underbrace{(I_3 - m \otimes m) P}_{\text{part in plane (same)}} - \underbrace{(m \otimes m) P}_{\text{opposite}}$

$= (I_3 - 2 \cdot m \otimes m) P$

Problem 4 $\Phi(x) = Ax + b \rightarrow$ affine transformation

Homogeneous matrix of the affine transformation Φ^{-1}?

Proof:

$\exists \Phi^{-1} \Leftrightarrow A$ invertible.

$\Phi(x) = y \Rightarrow Ax + b = y \Rightarrow y - b = Ax \mid \cdot A^{-1}$

$\Rightarrow A^{-1}(y - b) = x$

$\Rightarrow A^{-1} y - A^{-1} b = x = \Phi^{-1}(y) = Mx + t$

$\Rightarrow M = A^{-1}$

$\Rightarrow t = -A^{-1} b$

...mogeneous matrix: $[\Phi] = \begin{bmatrix} A & b \\ 0 & 1 \end{bmatrix}$

$$\left[\begin{array}{cc|cc} A & b & 1 & 0 \\ 0 & 1 & 0 & 1 \end{array}\right] \overset{\frac{1}{A}R_1 \to R_1}{\sim} \left[\begin{array}{cc|cc} 1 & b/A & 1/A & 0 \\ 0 & 1 & 0 & 1 \end{array}\right] \xrightarrow{R_1 - b/A \cdot R_2 \to R_1} \left[\begin{array}{cc|cc} 1 & 0 & 1/A & -b/A \\ 0 & 1 & 0 & 1 \end{array}\right]$$

$$\underbrace{\begin{matrix} 1/A & -b/A \\ 0 & 1 \end{matrix}}_{[\Phi]^{-1}}$$

$\Rightarrow [\Phi]^{-1} = \begin{bmatrix} A^{-1} & -b \cdot A^{-1} \\ 0 & 1 \end{bmatrix}$

Inverse linear part: A^{-1}

Inverse translation: $-A^{-1} \cdot b$

Problem 5 | π plane in $\mathbb{E}^3$ – normal vector n

?

$\pi: \langle n, x \rangle = c$

$v \parallel n$

composition of the orthogonal reflection in π followed by a translation by v $\Rightarrow$ reflection in $\pi': \langle n, x\rangle = c + \frac{1}{2}\langle v, n\rangle$

$T_v \circ Ref^{\perp}_{\pi}$ = reflection in

Proof:

plane $\langle n, x\rangle = c \Rightarrow$ reflected point: $Ref^{\perp}_{\pi}(x) = x - 2 \cdot \frac{\langle n, x\rangle - c}{\langle n, n\rangle} n \xrightarrow{n = \text{unit}}$

$\Rightarrow Ref^{\perp}_{\pi}(x) = x - 2 \cdot (\langle n, x\rangle - c) \cdot n$

Translation by v: $T_v(x) = x + v$

$(T_v \circ Ref^{\perp}_{\pi})(x) = x - 2 \cdot (\langle n, x\rangle - c) \cdot n + v$.

$$\left.\begin{matrix} v \parallel n \Rightarrow v = \lambda n. \\ |n| = 1 \end{matrix}\right\} \Rightarrow \lambda = \langle v, n\rangle$$

$\Rightarrow \langle v, n\rangle = \langle \lambda n, n\rangle = \lambda \langle n, n\rangle = \lambda \cdot |n|^2 = \lambda$ $\quad |n| = 1$

$\Rightarrow v = \langle v, n\rangle \cdot n$

$$(T_v \circ Ref^{\perp}_{\pi})(x) = x - 2(\langle n, x\rangle - c) \cdot n + \langle v, n\rangle n$$
$$= x - 2\left(\langle n, x\rangle - c - \frac{1}{2}\langle v, n\rangle\right) n$$
$$= x - 2\left(\langle n, x\rangle - \left(c + \frac{1}{2}\langle v, n\rangle\right)\right) n$$

?

$\pi': \langle n, x\rangle = c + \frac{1}{2}\langle v, n\rangle$

Problem 6 ✓ $\Phi(x) = Ax + b$ isometry $\Leftrightarrow$ $A^{-1} = A^T$. (A orthogonal)

Proof:

An isometry preserves distances $d(\Phi(P), \Phi(Q)) = d(P,Q)$

$\Phi(P) - \Phi(Q) = (AP+b) - (AQ+b) = A(P-Q) = P-Q$

$\Rightarrow$ translation b does not matter.

„$\Rightarrow$" Assume Φ isometry.

vector $u = P - Q \Rightarrow |Au| = |u| \quad |^2$

$\Rightarrow |Au|^2 = |u|^2$

$\Rightarrow \langle Au, Au \rangle = \langle u, u \rangle$

$\Rightarrow (Au)^T \cdot (Au) = u^T u$

$\Rightarrow u^T A^T \cdot (Au) = u^T u$

$\Rightarrow u^T A^T \cdot (Au) - u^T u = 0$

$\Rightarrow u^T (A^T \cdot (Au) - u) = 0$

$\Rightarrow u^T \underbrace{(A^T A - I)}_{} u = 0.$

$\Rightarrow A^T A = I \Rightarrow A^{-1} = A^T.$

„$\Leftarrow$" Assume $A^{-1} = A^T$

$\Rightarrow A^T A = I$

$(\forall)\ P, Q \Rightarrow d(\Phi(P), \Phi(Q)) = |A(P-Q)| \quad |^2$

$\Leftrightarrow |A(P-Q)|^2 = (P-Q)^T \underbrace{A^T A}_{I} (P-Q)$?

$\parallel$

$(A(P-Q))^2 \cdot (A(P-Q))$

$= (P-Q)^T (P-Q)$

$= |P-Q|^2$?

$= d(P,Q)$

$\Rightarrow d(\Phi(P) - \Phi(Q)) = d(P,Q)$

$\Rightarrow \Phi$ isometry.

Squared Euclidean norm:

$\|Av\|^2 = (Av)^T (Av)$

~~$A^T A =$~~ $v^T \cdot v = \langle v, v \rangle = \|v\|^2$

Problem 7 $A \in SO(2) \Leftrightarrow A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$, $\theta \in \mathbb{R}$.

A is an isometry $\Leftrightarrow A^T A = I$, orientation is preserved when det > 0.

$SO(2) = \{A \in M_2(\mathbb{R}) : A^T A = I, \det(A) = 1\}$ → columns are ~~orthogonal~~ orthonormal
→ det = 1

Proof:

"⇒" $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $A^T A = I$.

⇒ orthonormal columns ⇒ first column: $a^2 + c^2 = 1$ (length 1)

⇒ $\exists \theta \in \mathbb{R}$ s.t. $\begin{cases} a = \cos\theta \\ c = \sin\theta \end{cases}$ ⇒ first column: $\begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$

⇒ second column: - ⊥ to first column
- length 1

⇒ perpendicular vector: $\begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}$ or $\begin{pmatrix} \sin\theta \\ -\cos\theta \end{pmatrix}$

det A = 1 ⇒ preserve orientation: $\begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}$

$\Rightarrow A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

"⇐" $A^T A = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$. (1)

$\det A = \cos^2\theta + \sin^2\theta = 1$. (2)

(1), (2) ⇒ $A \in SO(2) \Leftrightarrow A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

Problem 8 Show that a direct isometry of $\mathbb{E}^2$ that fixes a point is either
- identity OR
- rotation

Proof.

$\Phi(x) = Ax + b$ affine isometry ⇒ $A^{-1} = A^T$ ⇒ A orthogonal. (rotation matrix)

direct isometry ⇒ det A = +1. ⇒ $A \in SO(2)$. ⇒ $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

Fix a point: $\Phi(P_0) = P_0$.

⇒ $A.P_0 + b = P_0$ ⇒ $b = P_0 - AP_0$

$\Phi(x) = Ax + (P_0 - AP_0) = P_0 + A(x - P_0)$

$\Phi(x) = P_0 + A(x - P_0)$ $\Rightarrow$ rotation centered at P_0. $\Rightarrow$ ROTATION!

If $\theta = 0 \Rightarrow A = I_2$, $\Phi(x) = P_0 + I_2(x - P_0) = I_2 \cdot x = x$.
(rotation angle)

$\Rightarrow$ IDENTITY. (= ROTATION with angle $\theta = 0$)

$\boxed{\text{Problem 9}}$ $Rot_{\theta_1, C_1} = f$, $Rot_{\theta_2, C_2} = g$ rotations with angles θ_1, θ_2 and centers C_1, C_2.

composition of two rotations $\Rightarrow$ translation $\Leftrightarrow$ $\theta_1 + \theta_2 = 2k\pi$

Rotation centered at C: $Rot_{\theta, C}(x) = C + R_\theta(x - C)$

$$= \underbrace{R_\theta}_{\text{lin. part} = R_\theta} x + (C - R_\theta C)$$

Proof:

$$f(x) = R_{\theta_1}x + \overbrace{(C_1 - R_{\theta_1}C_1)}^{b_1} = R_{\theta_1}x + b_1$$
$$g(x) = R_{\theta_2}x + \underbrace{(C_2 - R_{\theta_2}C_2)}_{b_2} = R_{\theta_2}x + b_2$$

$$(g \circ f)(x) = g(f(x)) = R_{\theta_2}(R_{\theta_1}x + b_1) + b_2 = \underbrace{R_{\theta_2}R_{\theta_1}}_{\text{lin. part}}x + R_{\theta_2}b_1 + b_2$$

$$R_{\theta_2}R_{\theta_1} = R_{\theta_1 + \theta_2}$$

$\boxed{\text{Translation: } x \mapsto x + t \Rightarrow \text{lin. part: } I_2}$

$g \circ f$ translation $\Leftrightarrow$ $R_{\theta_2}R_{\theta_1} = \text{lin. part} = R_{\theta_1 + \theta_2} = I_2$.
(possibly trivial)

$\Leftrightarrow$ $\theta_1 + \theta_2 = 2k\pi$, $k \in \mathbb{Z}$

↓

the translation vector may be 0 $\Rightarrow$ composition = identity.

$\boxed{\text{Problem 10}}$ $C(c_1, c_2)$ point. Homogeneous matrix of a rotation with angle θ and center C.

Rotation around C:
1) translate C into origin
2) rotate around origin
3) translate back

$$Rot_{\theta, C}(x) = C + R_\theta(x - C)$$

$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ rotation matrix around origin.

$x = \begin{pmatrix} x \\ y \end{pmatrix}, \quad c = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$

$$Rot_{\theta,c}(x) = c + R_\theta(x-c) = \underbrace{R_\theta x}_{\text{lin. part}} + \underbrace{(c - R_\theta c)}_{\text{translation part}}$$

$\Rightarrow A = R_\theta$

$\Rightarrow b = c - R_\theta c$

$$R_\theta c = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} c_1 \cdot \cos\theta - c_2 \cdot \sin\theta \\ c_1 \cdot \sin\theta + c_2 \cdot \cos\theta \end{pmatrix}$$

$$b = c - R_\theta c = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} - \begin{pmatrix} c_1 \cdots \\ c_1 \cdots \end{pmatrix} = \begin{pmatrix} c_1 \cdot (1-\cos\theta) + c_2 \cdot \sin\theta \\ c_2 \cdot (1-\cos\theta) - c_1 \cdot \sin\theta \end{pmatrix}$$

| Homogeneous matrix : $\Phi(x) = Ax + b \Rightarrow [\Phi] = \begin{bmatrix} A & b \\ 0 & 1 \end{bmatrix}$ |

Our case: Homog. matrix $= \begin{bmatrix} \cos\theta & -\sin\theta & c_1(1-\cos\theta) + c_2 \cdot \sin\theta \\ \sin\theta & \cos\theta & c_2(1-\cos\theta) - c_1 \cdot \sin\theta \\ 0 & 0 & 1 \end{bmatrix}$

$A = R_\theta$

$b = c - R_\theta c$

? $\boxed{\text{Problem 11}}$ Rotation with angle θ around origin after a reflection in the x-axis is a reflection in the line $y = \tan(\theta/2)x$.

→ Reflection in the x-axis

$(x,y) \mapsto (x,-y)$

$S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

→ Rotation by angle θ around origin:

$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

→ Rotation after reflection:

$$R_\theta \circ S = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$$

→ Reflection in a line through origin making angle θ with x-axis:

Reflection matrix $= \begin{pmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix} = R_\theta \circ S \quad (\Leftrightarrow) \ \theta = 2\alpha \Rightarrow \alpha = \frac{\theta}{2}$

→ Equation of the reflection line

$y = tg(\alpha) \cdot x$ → line through the origin making angle α with x-ax

$\alpha = \frac{\theta}{2}$

$\Rightarrow y = tg\left(\frac{\theta}{2}\right) x$

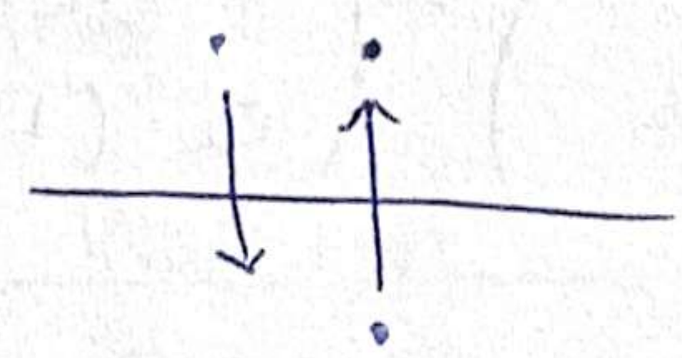
$\boxed{\text{Problem 12}}$ In dim 2, discuss the isometries obtained by composing a reflec in a line l_1 with a reflection in a line l_2. in terms of relative pos. of two

$\mathbb{E}^2$; l_1, l_2

$Ref^{\perp}_{l_1} \circ Ref^{\perp}_{l_2}$ in terms of rel. pos. of l_1, l_2.
(orthogonal reflections)

Case I: $l_1 = l_2$

$\phi = Ref^{\perp}_{l_1} \circ Ref^{\perp}_{l_2}$

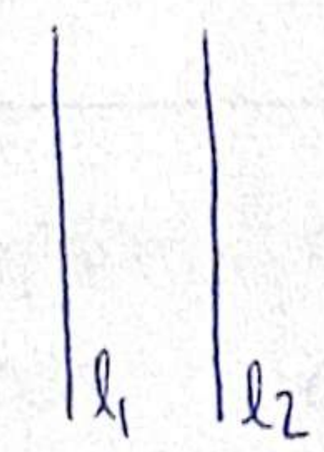


Intuitively, $\phi =$ identity

Case II: $l_1 \parallel l_2$

$\phi(x) = Ax + b = Ref^{\perp}_{l_1}$

$\psi(x) = Cx + d = Ref^{\perp}_{l_2}$



$A = C$

$Ref^{\perp}_{l_2} \circ Ref^{\perp}_{l_1} = \psi(\phi(x)) = C(Ax+b) + d = \underbrace{CA}_{\text{lin. part.}}x + \underbrace{Cb+d}_{\text{translation part.}}$

$\Rightarrow \psi \circ \phi$ is direct isometry since $\det A = -1 = \det C$.

identity, translation, rotation

identity
$\begin{cases} CA = I_2 \\ Cb + d = 0 \end{cases}$
$\Updownarrow$

$\left.\begin{array}{l} C = A^{-1} \\ A = C \end{array}\right\} \Rightarrow A^{-1} = A = C$

$A^{-1} = A$ because they are reflections

$A = C$ because $\phi = \psi$

?

$\Rightarrow \det AC = \det(A) \cdot \det(C) = (-1) \cdot (-1) = 1$.

$\left.\begin{array}{l} Cb + d = 0 \\ A = C \end{array}\right\} \Rightarrow Ab + b = 0$. because $(\psi \circ \phi)(x) = CA + Cb + d = x + \underbrace{Cb}_{-b} + d$ (⊥ reflection)

?

$\Rightarrow Ab = -b$ because $\boxed{b \perp \text{axis of } \psi}$

$Cb + d = 0 \Rightarrow Cb = -d = -b$

?

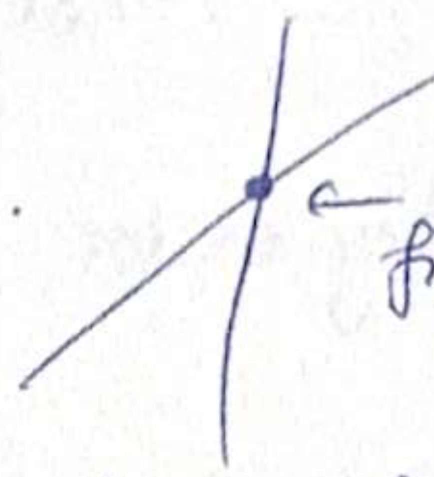
$\Rightarrow$ translation with $d - b$.

$x) = Ax+b$ reflection $\iff$ $\boxed{b \perp \text{axis of } \varphi}$

$\to$ because we want reflection, not glide reflection.

Case III: $l_1 \cap l_2 = 1$ point

$\varphi \circ \phi$ direct isometry with one fixed point $\Rightarrow$ ROTATION.



fixed.

$\boxed{\text{Problem 13}}$ Show that $A \in SO(3)$ admits 1 as eigenvalue.

A = orthogonal matrix if $A^T A = I \Rightarrow A^{-1} = A^T$. $\Rightarrow O(n)$ set of orthogonal matrices.

A = special orthogonal matrix if $\begin{cases} A \in O(n) \\ \det A = 1 \end{cases} \Rightarrow SO(n) = \{A \in O(n) : \det A = 1\}$

$A \in SO(3) \Rightarrow \begin{cases} A^T A = I_3 \\ \det A = 1 \end{cases}$ admits $\lambda = 1 \iff$

$\iff \det(A - \lambda I_3) = 0$, $\lambda = 1$

$\det(A - I_3) = 0 \iff \begin{vmatrix} \dots \end{vmatrix} = 0$

$\boxed{\det A = \det A^T = 1}$, $\boxed{\det A \cdot \det B = \det(A \cdot B)}$

$$\det(A - I_3) = \det(A^T) \cdot \det(A - I_3) = \det(A^T(A - I_3)) =$$

$$= \det(\underbrace{A^T A}_{\substack{I_3 \\ \text{(orthogonal)}}} - A^T) = \det(I_3 - A^T) = \det(I_3 - A)^T = \det(I_3 - A)$$

$\Rightarrow \det(A - I_3) = \det(I_3 - A) = -\det(A - I_3)$

~~$\iff A - I_3 = I_3 - A$~~ $\Rightarrow 2\det(A - I_3) = 0 \Rightarrow \det(A - I_3) = 0$.

$\Rightarrow \lambda = 1$ eigenvalue, indeed

~~$\det(I_3 - A) = \det(-(A - I_3)) = (-1)^3 \cdot \det(A - I_3)$~~

$\boxed{\text{Problem 14}}$ Show that $A \in O(3)$ with $\det(A) = -1$ admits -1 as eigenvalue.

$\lambda = -1$ eigenvalue $\iff \det(A - (-1)I) = 0 \iff \det(A + I) = 0$.

$A \in O(3) \Rightarrow \begin{cases} A^T A = I \\ \det A = -1 = \det(A^T) \end{cases}$

$$\underline{-\det(A + I)} = \det(A^T) \cdot \det(A + I) = \det(A^T(A + I)) = \det(\underbrace{A^T A}_{I} + A^T) =$$

$$= \det(I + A^T) = \det((I + A)^T) = \underline{\det(I + A)}$$

~~$I + A = A + I \mid \det$~~

$\Rightarrow -\det(I + A) = \det(A + I) = \det(I + A)$

$\Rightarrow 2\det(A + I) = 0 \Rightarrow \det(A + I) = 0$

$\Rightarrow -1$ eigenvalue, indeed.

Problem 15 v = unit vector $\Rightarrow |v|=1$

$\theta \in \mathbb{R}$

Show that rotation with angle θ and axis $\mathbb{R}v$ is given by:

$$Rot_{v,\theta}(x) = \cos(\theta)x + \sin(\theta)(v \times x) + (1-\cos\theta)\langle v, x\rangle v$$

Any vector $x \in \mathbb{E}^3$ can be split into: $x = x_{\parallel} + x_{\perp}$

$x_{\parallel} = \langle v, x\rangle v$

$v \times x \perp v$ $\quad x_{\perp} = x - \langle v, x\rangle v$

$\perp x$

$Rot_{v,\theta}(\langle v, x\rangle v) = \langle v, x\rangle v = Rot_{v,\theta}(x_{\parallel})$

$x_{\perp} = x - \langle v, x\rangle v$

$Rot_{v,\theta}(x_{\perp}) = \cos\theta\, x_{\perp} + \sin\theta\,(v \times x)$

$Rot_{v,\theta}(x) = Rot_{v,\theta}(x_{\parallel}) + Rot_{v,\theta}(x_{\perp})$

$= \langle v, x\rangle v + \cos\theta\, x_{\perp} + \sin\theta\,(v \times x)$

$x_{\perp} = x - \langle v, x\rangle v$

$\Rightarrow Rot_{v,\theta}(x) = \langle v, x\rangle v + \cos\theta\,(x - \langle v, x\rangle v) + \sin\theta\,(v \times x)$

$= \langle v, x\rangle v + \cos\theta\, x - \cos\theta\,\langle v, x\rangle v + \sin\theta\,(v \times x)$

$= \cos\theta\, x + \sin\theta\,(v \times x) + (1-\cos\theta)\langle v, x\rangle v$

Problem 16 Deduce can. eq. of the ellipse.

$\mathcal{E}: \{P \in \mathbb{E}^2 \mid d(P,F_1) + d(P,F_2) = 2a\},\ (\forall)\, a>0.$

$$\boxed{\mathcal{E}: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1}$$

$u \to 4$

Proof:

In $\mathcal{K}$, $F_1(c,0), F_2(-c,0)$, $c>0$, $P(x,y)$

$\mathcal{E}: \underbrace{\sqrt{(x-c)^2+(y-0)^2}}_{d(P,F_1)} + \underbrace{\sqrt{(x+c)^2+(y-0)^2}}_{d(P,F_2)} = 2a$

$\Rightarrow \sqrt{(x-c)^2+y^2} + \sqrt{(x+c)^2+y^2} = 2a$

$\Rightarrow \sqrt{(x-c)^2+y^2} - 2a = -\sqrt{(x+c)^2+y^2} \quad |^2$

$\Rightarrow (x-c)^2 + \cancel{y^2} - 4a\sqrt{(x-c)^2+y^2} + 4a^2 = (x+c)^2 + \cancel{y^2}$

$\Rightarrow -2xc - 4a\sqrt{(x-c)^2+y^2} + 4a^2 = 2xc$

$\Rightarrow 4a^2 - 4xc - 4a\sqrt{(x-c)^2+y^2} = 0 \quad |:4$

$\Rightarrow a^2 - cx = a\cdot\sqrt{(x-c)^2+y^2} \quad |^2$

$$a^4 - 2a^2cx + c^2x^2 = a^2(x^2 - 2cx + c^2 + y^2)$$

$$\Rightarrow a^4 + c^2x^2 = a^2x^2 + c^2a^2 + a^2y^2 \quad /: a^2$$

$$\Rightarrow a^2 + \frac{c^2x^2}{a^2} = x^2 + c^2 + y^2$$

$$\Rightarrow \underbrace{a^2 - c^2}_{b^2} = x^2 + y^2 - \frac{c^2x^2}{a^2} \quad |: b^2 = a^2 - c^2$$

$$\Rightarrow 1 = \frac{x^2}{b^2} + \frac{y^2}{b^2} - \frac{c^2x^2}{b^2a^2}$$

$$\Rightarrow 1 = \frac{x^2}{b^2}\left(1 - \frac{c^2}{a^2}\right) + \frac{y^2}{b^2}$$

$$\Rightarrow 1 = \frac{x^2}{\cancel{b^2}} \cdot \frac{\cancel{a^2 - c^2}}{a^2} + \frac{y^2}{b^2}$$

$$\Rightarrow 1 = \frac{x^2}{a^2} + \frac{y^2}{b^2} : \mathcal{E}$$

$\boxed{\text{Problem 17}}$ $\mathcal{E}: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ Equations for the tangent lines of a given slope

$$(S)\begin{cases} \mathcal{E}: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \\ l: kx + m = y \end{cases} \Rightarrow \frac{x^2}{a^2} + \frac{(kx+m)^2}{b^2} = 1$$

(S) has solutions $\Leftrightarrow \underbrace{(b^2 + a^2k^2)}x^2 + \underbrace{2kma^2}x + \underbrace{a^2(m^2 - b^2)} = 0$ has sol.

$\Leftrightarrow \Delta \geq 0.$

$\Delta > 0 \Rightarrow 2 \neq$ solutions $\Rightarrow$ line $\cap$ ellipse in 2 points.

$\Delta = 0 \Rightarrow 2 =$ solutions $\Rightarrow$ tangent

$$\Delta = \underbrace{4a^2b^2}_{\substack{>0 \\ \neq 0}}(a^2k^2 + b^2 - m^2)$$

Case I: $\Delta < 0$

no intersection

Case II: $\Delta > 0 \Rightarrow a^2k^2 + b^2 - m^2 > 0$

$m \in (-\sqrt{a^2k^2 + b^2}, \sqrt{a^2k^2 + b^2})$

$\Rightarrow$ intersection in 2 points

Case III: $\Delta = 0 \Rightarrow a^2k^2 + b^2 - m^2 = 0$

$\Rightarrow m = \pm\sqrt{a^2k^2 + b^2}$

If k is given (the slope) $\Rightarrow \boxed{y = kx \pm \sqrt{a^2k^2 + b^2}}$

$\boxed{\text{Problem 18}}$ ✓ $\mathcal{E}: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Equation for the tangent line at given (x_0, y_0) of the ellipse.

$$\begin{cases} \mathcal{E}: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \\ \ell: \begin{cases} x = x_0 + t \cdot v_x \\ y = y_0 + t \cdot v_y \end{cases} \end{cases}$$

ℓ is tangent to $\mathcal{E}$ when $\ell \cap \mathcal{E}$ in <u>one</u> point.

$(\Leftrightarrow) \; |\ell \cap \mathcal{E}| = 1.$

$$\frac{(x_0 + t \cdot v_x)^2}{a^2} + \frac{(y_0 + t \cdot v_y)^2}{b^2} = 1, \text{ in terms of } t.$$

$$\Rightarrow \frac{x_0^2 + 2t \cdot x_0 \cdot v_x + t^2 \cdot v_x^2}{a^2} + \frac{y_0^2 + 2 \cdot t \cdot y_0 \cdot v_y + t^2 v_y^2}{b^2} = 1$$

$$\Rightarrow t^2 \left(\frac{v_x^2}{a^2} + \frac{v_y^2}{b^2} \right) + 2t \cdot \left(\frac{x_0 \cdot v_x}{a^2} + \frac{y_0 \cdot v_y}{b^2} \right) + \overbrace{\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} - 1}^{0} = 0$$

Take $t = 0$ because (x_0, y_0) corresponds to $t = 0$.

this coefficient $\frac{x_0 v_x}{a^2} + \frac{y_0 v_y}{b^2}$ has to be 0 in order for $t = 0$ to be a double solution ($\Rightarrow$ double intersection point (2 equal points)).

$(\Leftrightarrow) \; \langle n, v \rangle = 0$

$(\Leftrightarrow) \; n\left(\frac{x_0}{a^2}, \frac{y_0}{b^2} \right)$ normal vector for ℓ.

$(\Leftrightarrow) \; \ell: \frac{x_0}{a^2}(x - x_0) + \frac{y_0}{b^2}(y - y_0) = 0.$

$$\ell: \frac{x_0 x}{a^2} - \frac{x_0^2}{a^2} + \frac{y_0 y}{b^2} - \frac{y_0^2}{b^2} = 0$$

$$\ell: \frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} \underbrace{- \frac{x_0^2}{a^2} - \frac{y_0^2}{b^2}}_{-1 \text{ since } (x_0, y_0) \in \mathcal{E}_{a,b}} = 0$$

$$\Rightarrow \mathcal{T}_{(x_0, y_0)} \mathcal{E}_{a,b}: \frac{x x_0}{a^2} + \frac{y y_0}{b^2} = 1$$